



2024

Public

Interest rate risk arising from non- trading book activities (IRRBB)

Tamás Zoltán, Corporate Treasury Senior Manager

Course Contents

1 IRRBB Framework

2 NII – Net Interest Income

3 EVE – Economic Value of Equity

4 IRRBB Management at Citi

5 Fund Transfer Pricing Principles

IRRBB Framework

Introduction

Interest rate risk arising from non-trading book activities (IRRBB) is an important financial risk for credit institutions, which is considered under Pillar 2.

What is IRRBB?

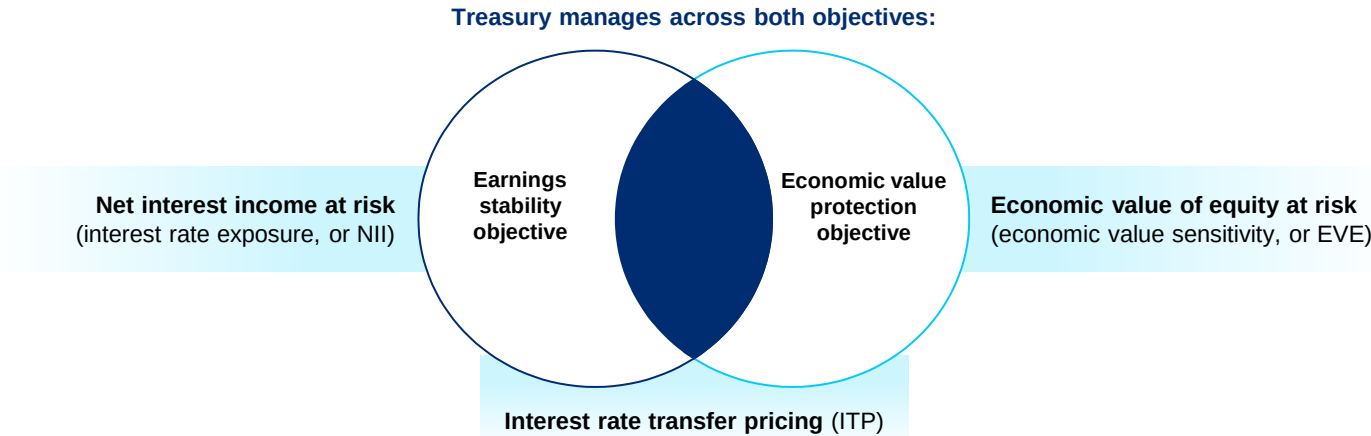
- Changes in interest rates can affect a bank's earnings and its Economic Value (Present Value).
- Interest Rate Risk in the Banking Book covers specifically all assets and liabilities which generate accrual P&L.
- IRRBB does not cover assets and liabilities which create Marked To Market P&L

How is IRRBB Measured?

- Interest Rate Exposure ("NII"). This is a model with a 1 year horizon that calculates how accrual P&L will behave across the balance sheet under parallel yield curve shocks.
- Economic Value Sensitivity ("EVE"). This is a model which calculates changes in the Economic Value (Present Value) of the balance sheet under parallel yield curve shocks.
- DV01 is the Dollar Value change in value from a 1bps change in rates. It is used to calculate the risk run in AFS positions as well as Markets Treasury total P&L.

Objectives and Principles of IRR Management

IR Risk Metrics ensure the short and long term risks from IRRBB to earnings and capital are captured



Overview of Interest Rate Risk in the Banking Book (IRRBB)

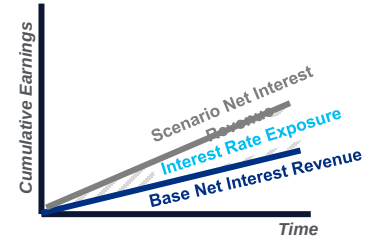
1. IRRBB arises through the normal course of business activities (e.g., deposits and lending)
2. IR Risk Metrics ensure the short and long term risks from IRRBB to earnings and capital are captured
3. Net Interest Income (NII) captures impact on earnings
4. Economic Value Sensitivity (EVE) captures longer term impact on value (i.e., Economic Value of Equity, or EVE)
5. OCI DV01 is also used as a metric to manage the risk to capital from AFS OCI
6. Treasury manages IRRBB through ITP while adhering to limits aligned to the firm's Risk Appetite Framework

IRRBB Framework

Interest Rate Risk in the Banking Book is reported using short term (NII) and long term (EVE) measures, and managed on a daily basis via the Interest Rate Transfer Pricing (ITP) process

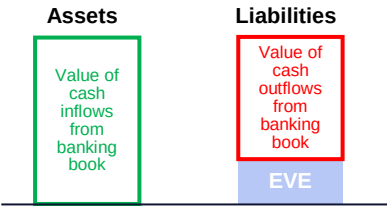
IRRBB Measurement Metrics
(Required by regulation)

Net interest income at risk (interest rate exposure, or NII)



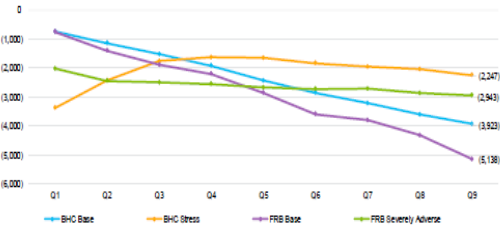
- Measures near term **risks to the firm's earnings**:
 - Pre-tax impact on Net Interest Revenue due to defined shifts in interest rates
 - Forecasted over the near term (1 or 2 years); measured monthly

Economic value of equity at risk (economic value equity, or EVE)



- Measures **risks to long-term franchise value**:
 - Changes in economic value due to interest rate movement
 - Forecasted over full life (contractual or via assumption); measured monthly

OCI DV01 (AFS OCI capital at risk on DV01 basis)



- Measures **interest rate risk component of firms CET1 capital at risk**:
 - Changes in value impacting OCI due to interest rate movement
 - Forecasted stress testing approach to measure losses; measured monthly

IRRBB Management Process

Lending Unit

ITP funding at risk-free rate

- Offsetting interest rate risk

Deposit Unit

ITP investment at risk-free rate:

- Offsetting interest rate risk

Interest rate transfer pricing (ITP)

- Transfers to Treasury:
 - Funding needs
 - Investment needs
 - Interest rate hedging profile
- Minimizes residual interest rate risk in businesses, subject to balance sheet constraints

Treasury

- Manages net:
 - Interest rate risk (i.e., NII, EVE)
 - Investment portfolio (i.e. OCI)
- Manages daily DV01 metrics supporting Treasury and risk management
- Limits set and governed by ALCO

1.Current internal NII metric anchored on accrual earnings / Net Interest Revenue (NIR)

2.Economic Value of Equity (EVE) is net economic value of assets, net of economic value of liabilities, using present value calculations that discount cash flows derived from current positions

3.Subject to balance sheet constraints

Measurement Metrics

NII

- Interest Rate Exposure (NII) is used to measure the change in forecasted interest income and expense due to changing interest rates
- Standard scenarios are 100bp parallel upward and downward moves on the rate curve
- NII is calculated by using balances over a 1- Year period and applying different rate scenarios to the cashflows over the 1-Year period

EVE

- Economic Value of Equity (EVE) metric is used to measure the change in economic value due to changing interest rates
- Economic Value of Equity (EVE) is the difference in the present value between total assets and total liabilities. It is calculated by discounting cash flows over the lifetime of the products
- Standard scenarios are 100bp parallel upward and downward moves on the rate curve
- EVE is a delta between the base EVE and EVEs calculated with two different rate scenarios applied to the curves.

DV01

- In addition to EVE, DV01 is used to measure the change in market value of interest-bearing products in case of a 1bp change in the underlying yield curves.
- DV01 is applied to products held in both trading and banking book as opposed to EVE.
- DV01 is commonly used by Markets Treasury to measure the outstanding risk managed by them

NII – Interest Rate Exposure

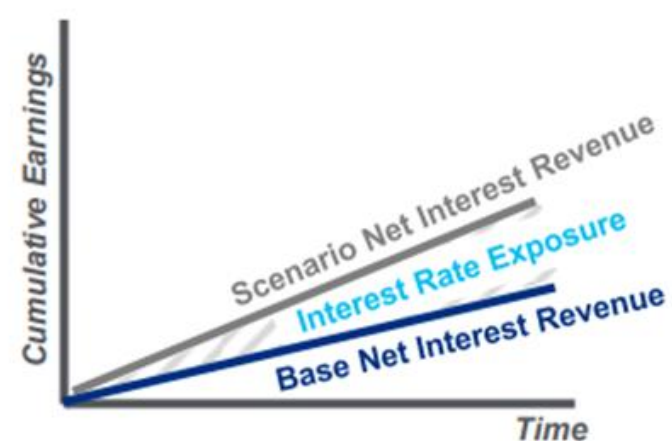
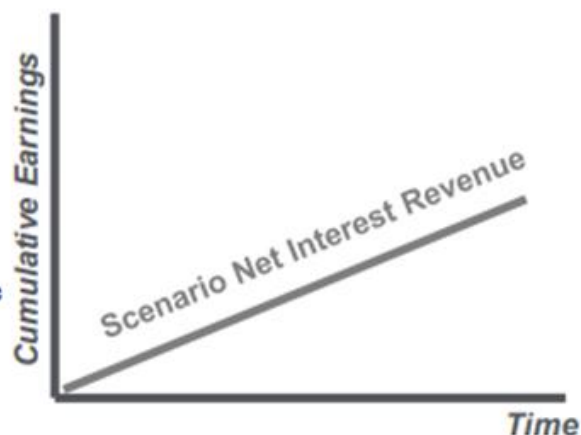
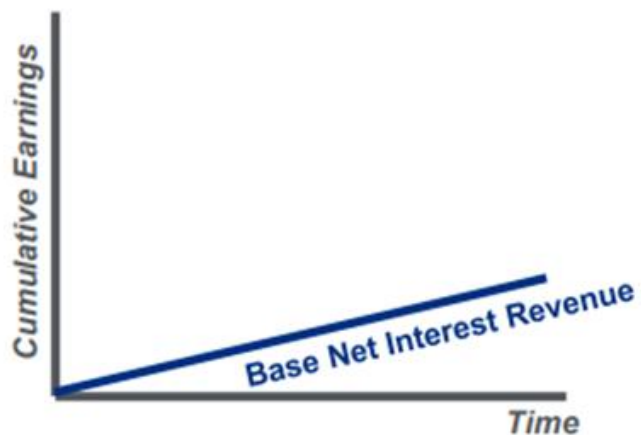
NII- Interest Rate Exposure

Metric	Definition	Formula	Calculation Methodology
NII	NII is an income metric which measures potential <u>accounting earnings</u> (accrual) as a result of interest rate exposure in the banking book.	$\text{NIR} = \text{Interest Payments on Assets} + \text{Interest Payments on Liabilities}$	Base Scenario: NIR Shock Scenario: NIR Shock NIR-Base NIR = NII

Model forecasted Net Interest Revenue

Recalculate forecast under shock to rates

Compute difference between forecasts



IRE calculated over 1 year, 2 year etc.

NII- Interest Rate Exposure

NII addresses potential changes in Net Interest Revenue (Accounting Earnings) from changes in interest rates.

Term	Key Drivers	Computation	Flooring
• Captures Short Term IRR • 12-24 months into the future	• Average Balance • Rate Shock Size • Tenor and Rate Reset Frequency • Assets • Liabilities • Deposit Beta • Change in customer rate relative to change in market rate • Implied Beta • For non-deposit accounts • Inverse relationship to time until next rate reset date.	• Shocked Scenario Net Interest Revenue (NIR)-Base Scenario NIR • $NII \sim AVG \text{ Bal} * \text{Beta} * \text{Rate Shock}$ • NII is aggregated across balance sheet lines (Assets-Liabilities)	• When yields are near 0%, large shocks down will experience “Beta Flooring” • Yield in Down Scenario is “floored” at 0%

NII- Interest Rate Exposure Example

Example:

- \$1000 Investment Average Balance, monthly coupon.
- 6% annual yield with 6mo repricing frequency (50% Beta)
- Rates Drop 100 basis points on M0 (Flat Yield Curve)

		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	Total
Base Scenario	Base Yield	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%
	Interest Revenue	5	5	5	5	5	5	5	5	5	5	5	5	60
Shocked Scenario	100bps DN	6%	6%	6%	6%	6%	6%	5%	5%	5%	5%	5%	5%	5.5%
	Interest Revenue	5	5	5	5	5	5	4.2	4.2	4.2	4.2	4.2	4.2	55
NII DN 100		0	0	0	0	0	0	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8	-5.0



Yield drops at the 6-month mark,
when Asset reprices

NII is the difference in Net
Interest Revenue

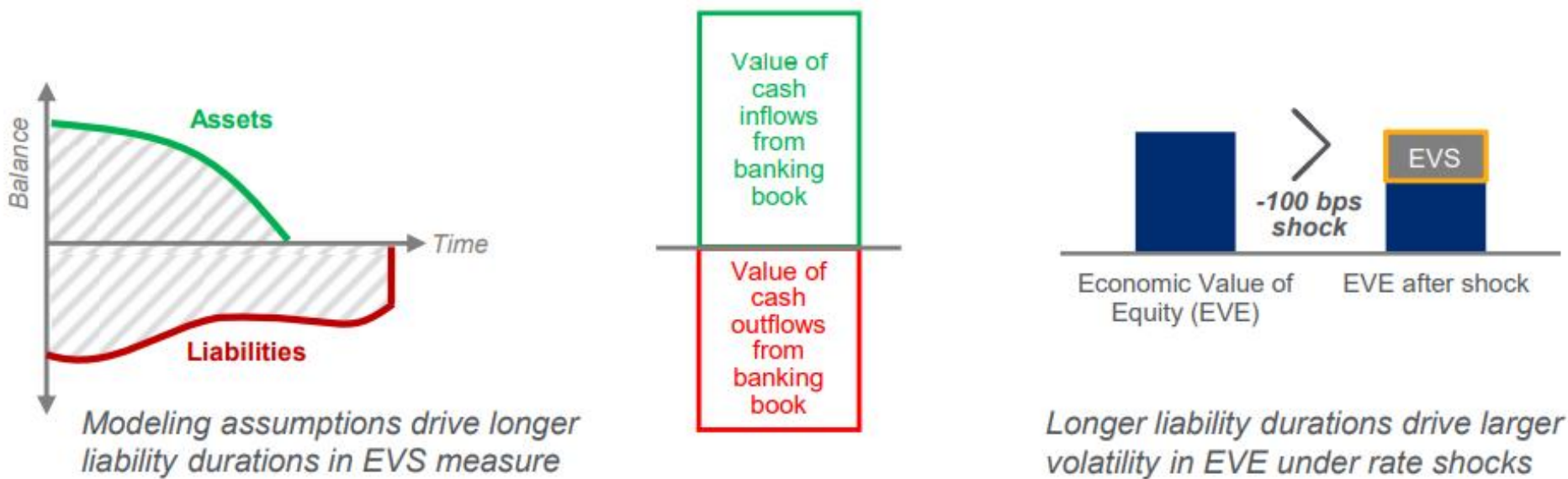
P&L impact from a rise in rates	
Assets	
Loans with short repricing	Strong pick up in P&L
Liabilities	
Term Deposits - short dated	Strong hit to P&L
NMD Deposits - Indexed	Strong hit to P&L
NMD Deposits - Managed	Weak hit to P&L depending on beta

- Overall, the rise in rates will create P&L on this balance sheet.
- This is because the assets are more sensitive to changes in P&L than the liabilities.
- In other words, the “beta” of the assets is higher than the beta of the liabilities.
- The key reason for this is that the interest rate rises are not passed on quickly to the Managed Non-defined maturity deposits, i.e. they have a lower beta.

EVE – Economic Value of Equity

EVE Visual Illustration

Metric	Definition	Formula	Calculation Methodology
EVE	EVE is an <u>economic</u> metric which measures the potential impact on capital as a result of interest rate exposure in the banking book.	$EVE = PV \text{ of Asset Cash Flows} - PV \text{ of Liability Cash Flows}$	Base Scenario: base EVE Shock Scenario: stressed EVE Shock EVE-Base EVE = delta EVE



EVE- Economic Value Sensitivity

EVE addresses economic (Present Value) changes in value of Balance Sheet Equity from changes in interest rates.

Term	Key Drivers	Computation	EVE
• Captures Long Term IRR	• Duration • Market Value of Assets/Liabilities • Discount Rates • Tenor and Rate Reset Frequency • Assets • Liabilities	• Shocked Scenario EVE-Base Scenario EVE • EVE is aggregated across balance sheet lines (Assets-Liabilities)	• EVE =Economic Value of Equity • Market Value of Assets-Market Value of Liabilities • Market Value = Present Value

EVE- Economic Value of Equity

Example

- Balance sheet consists of \$1000 in Market Value of Assets (fixed) and \$800 in Market Value of Liabilities (fixed). MV = Present Value
- Assets have an average Duration of 1 year
- Liabilities have an average Duration of 2 years
- Rates Drop 100bps

	Base Scenario	Shocked Scenario -100bps	EVE DN 100
Assets	1000	1010	+10
Liabilities	800	816	-16
EVE	<u>200</u>	<u>194</u>	<u>-6</u>

} Asset Market Value increased
less than Liability Market Value

Equity Value Decreased



Impact from a rise in rates on Economic Value	
Assets	
Loans with short repricing	Small fall in positive EV
Liabilities	
Term Deposits - short dated	Small fall in negative EV
NMD Deposits - Indexed	Small fall in negative EV
NMD Deposits - Managed	Strong fall in negative EV

- Overall, the rise in rates will create Economic Value on this balance sheet
- This is because the liabilities EV (negative) falls more than asset EV (positive).
- The key driver are the managed non-defined maturity deposits, which tend to have a longer duration than other balance sheet items. This is driven by their modelled run-off profile and beta.
- A lower beta means that changes in interest rates have a limited impact on the client rates, i.e. the effective duration of the liability is longer. The EV of assets and liabilities with longer effective duration are more sensitive to changes in interest rates.

Other IRRBB Concepts

Concept	Definition	Formula/Example	Calculation Methodology
DV01	Similar to EVE, DV01 is a value sensitivity measure which measures the potential impact in a change of value per 1 bps movement in interest rates .	$DV01 = \frac{(PV_{-1bp}) - (PV_{+1bp})}{2}$	Base Scenario: EVE Shock Scenario: EVE (1 bp shift) Shock EVE-Base EVE = DV01
OCI DV01	Value sensitivity within Other Comprehensive Income from changes in value in Available for Sale Securities in response to a 1 basis point change in interest rates.		Base Scenario: AFS MV Shock Scenario: AFS MV(1bp shift) Shock AFS MV-Base AFS MV= OCI DV01
Basis Risk	Risk to earnings or capital from interest rate movements due to changing rate relationships between different reference, currency and tenor curves.	±100 basis point shift NII, EVE.	NII and EVE are calculated based on shocks to different reference, currency and tenor curves. Shock-Base.
Curve Risk	Risk to capital from changes in the shape of the yield curve (Non-parallel shifts).	Bull/Bear Flattening/Steepening shifts.	Shock EVE-Base EVE for different non-parallel shifts.
Optionality Risk	Risk to earnings or capital arising from changes in value of assets/liabilities with embedded options in response to changes in interest rates.	Products with pre-payment options, such as mortgages and credit lines may experience shorter maturities if rates decrease, potentially decreasing interest earnings and capital value.	Prepayment risk is captured in Global Optionality Survey.
Global Systematic Stress Test (GSST)	Specific combinations of market and macroeconomic shocks designed to adversely impact the firm's financials. Scenarios may be historically based or hypothetical.	1 in 10 Business Specific Stress Tests	Ruby-Generated NII/EVE Scenarios based on historical or hypothetical inputs.

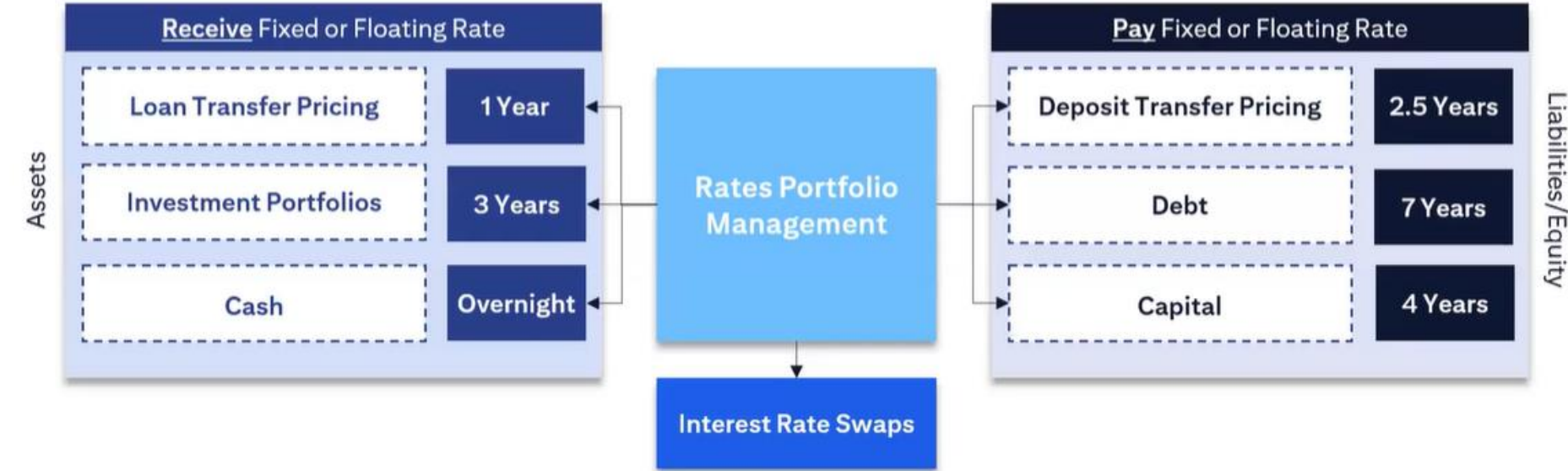
IRRBB Management at Citi

Interest Rate Positioning & Strategy

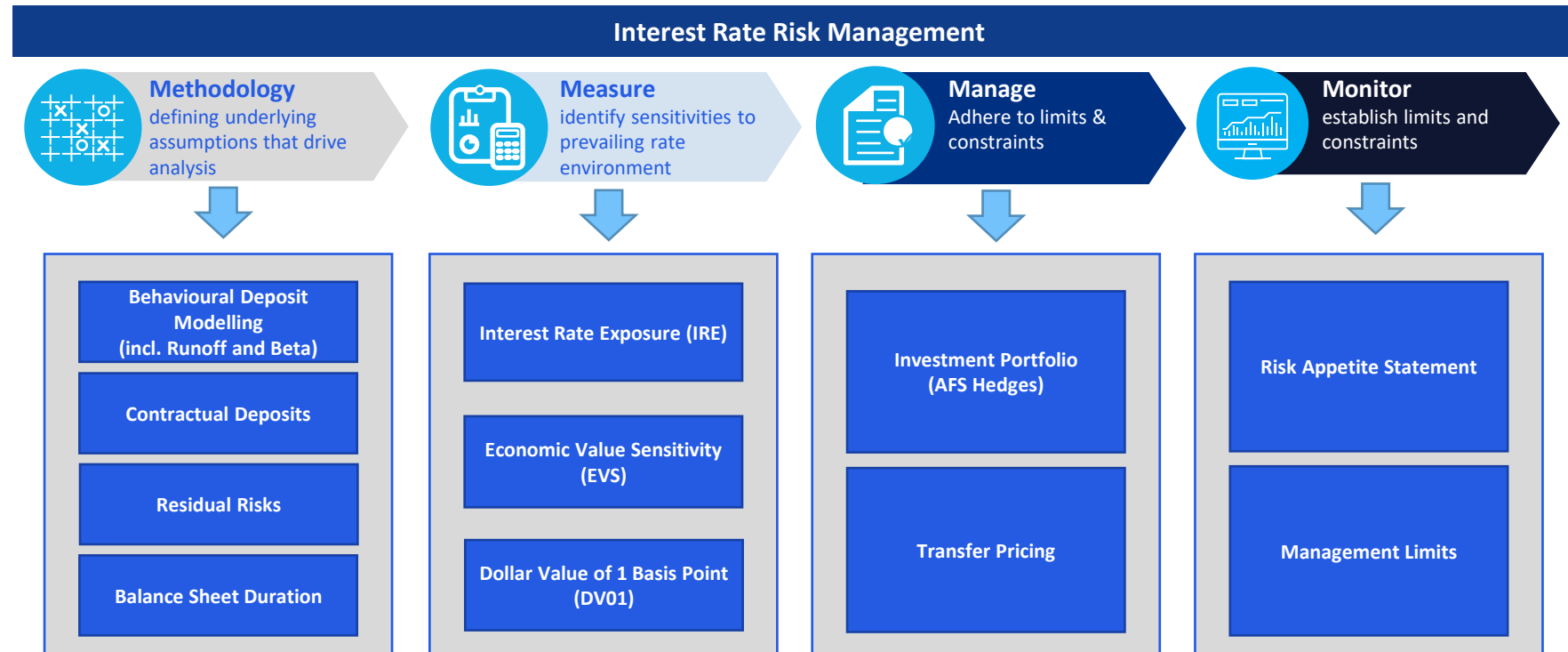


Interest Rate Risk*

Interest Rate Risk is created by differences in asset and liability maturities, benchmark rates and option characteristics



Overview



Treasury in general has three instruments to balance interest rate risk arising from banking book:

- **Buying AFS Securities**
- **Entering into interest rate derivative positions**
- **Operate a Hedge Accounting program**

IRRBB Management Tools

AFS Bonds

- AFS securities are in general fixed income securities (bonds). Purpose of these securities is to generate a steady interest income and extend duration on the asset side. These securities can be sold or used as collateral in repo transactions if needed.
- AFS securities are accounted for on a FVOCI basis.
- Buying long dated AFS will lower the beta of the asset side of the balance sheet and so hedge NII risk generated by lower beta deposits.
- Buying long dated AFS will increase the duration of the asset side of the balance sheet and so hedge EVE risk generated by longer duration deposits

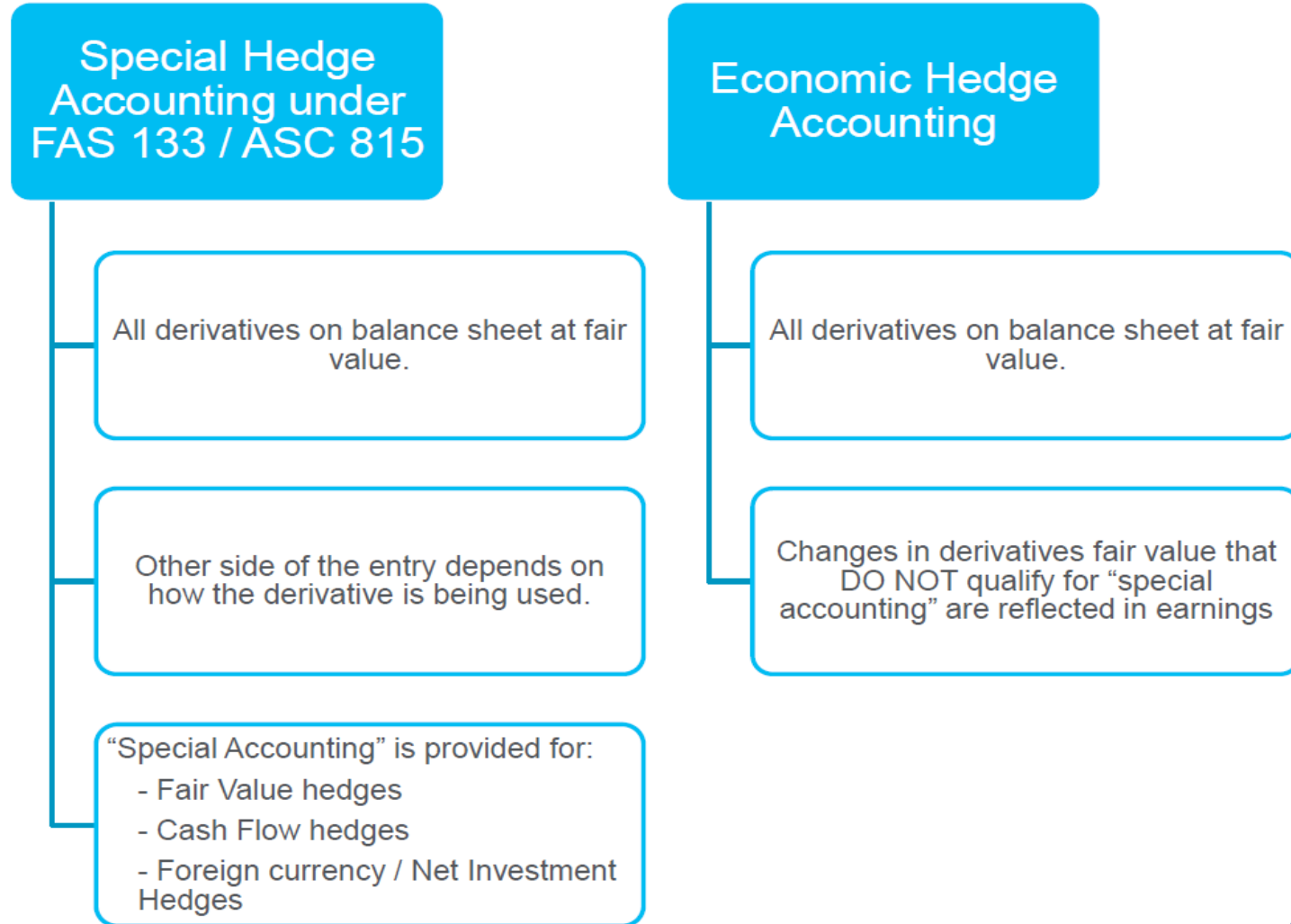
IR Derivatives

- Alternative to bond purchases, Treasury can trade interest rate swaps in which they exchange variable rate payments against a fixed rate to hedge the duration of liabilities.
- Interest rate swaps are generally accounted for on FVPNL basis against Mark-To-Market. This has no OCI implications but can generate PNL volatility.
- Fair-value hedging treatment could be applied, which is a complex accounting principle that can reduce the PNL impact of hedging transactions.
- In case the asset side duration is too long, it can also be reduced through the usage of IRS in which Markets receive a variable rate and pay a fixed rate. Same accounting principles apply.

Hedge Accounting

- For items under FVOCI treatment, the accrual interest of coupons, and the differential between par value and purchase price over time is booked against P&L.
- The difference between the accrual price and the real market price is accounted for against OCI. OCI is unrealized gain or loss and accounted for against equity.
- In case a bond gets sold prior to maturity, the accumulated OCI automatically moves under the P&L line.

Accounting Hedges vs Economic Hedges



Types of Hedge Accounting



- Hedged risk item's fair value will change in response to interest rates, foreign exchange rates, or market prices changes.
- Gains and losses on the hedging instrument and the offsetting gains and losses on the hedged item are both recognised in P&L.



- Hedged risk item's future cash flows will change in response to such variables.
- Gain or loss on the hedging instrument is initially recognised in OCI (equity) and subsequently recycled from equity to P&L as the hedged item affects P&L.



- Risk is that the carrying amount of a net foreign investment will change in response to FX rate movements.
- Gain or loss on the hedging instrument is initially recognised in OCI (equity) and subsequently recycled to P&L from equity on disposal of that foreign operation.

Example

Fair Value Hedges: Example

- **Hedged Item**

- Hedged Item:
- Notional: 100
- i fixed: 8%
- Type: AFS Security
- n= 5 years
- Mtm Value: 96

- **Hedging Derivative:**

- Notional: 100
- Type: IRS
- Pay i fixed: 8%
- Receive i float: TIIE 9%
- n= 5 years
- Mtm Value: 5

AFS Sec	
100	4
Mtm Der	
5	

Cash	
	100
P&L	
5	5

OCI	
4	5
	1

→ Credit Risk
other Risk

Fund Transfer Pricing Principles

Funds Transfer Pricing Principles

The general principal of FTP in regards to IRRBB is to transfer the interest rate risk from the businesses to Markets. The transfer price is the rate at which Treasury provides funding for an asset or makes an investment for a liability.

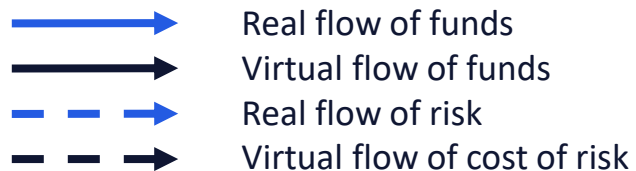
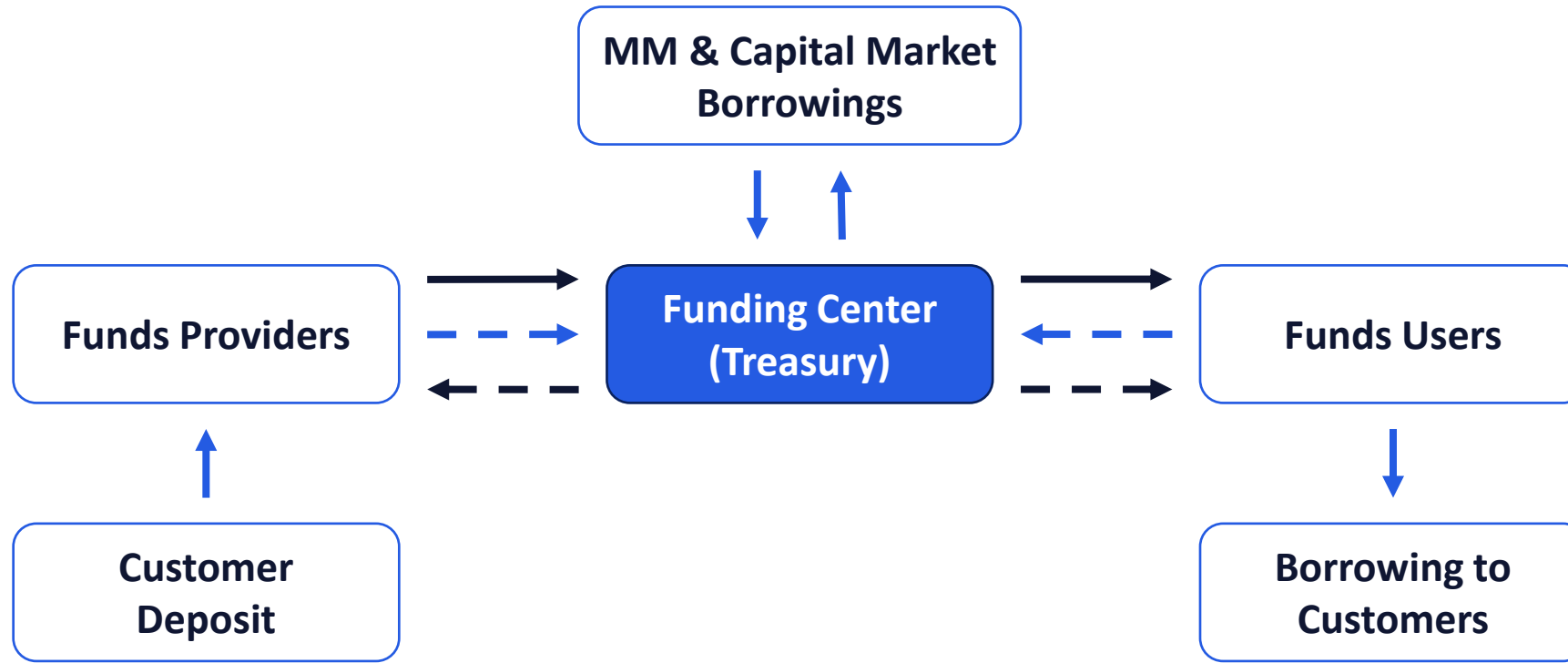
The goal of transfer pricing

1. Reflect true funding costs.
2. Separate risk elements (Credit Risk, Liquidity Risk, Interest Rate Risk, Operational Risk).
3. Allows management to assess profitability on a business-by-business basis.
4. Obtain correct profitability.

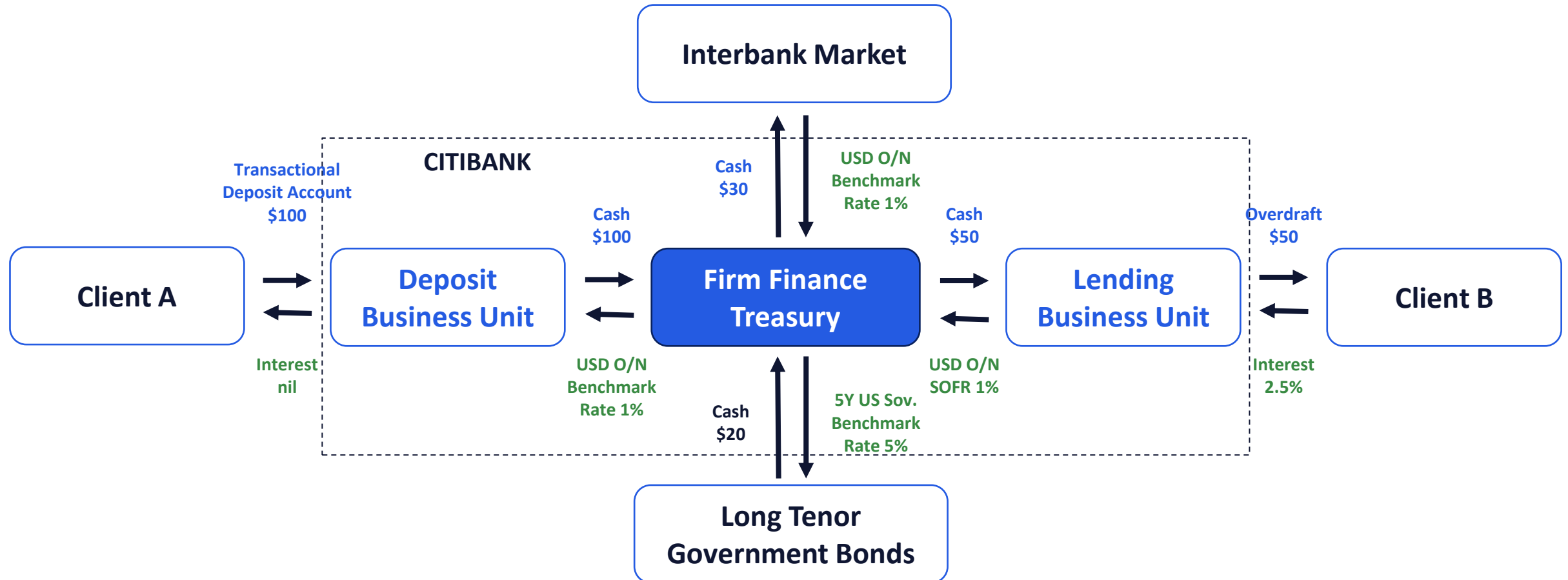
The concept of the appropriate transfer pricing

1. Value funding costs on an “arm’s length” basis.
2. Incentivize businesses to reduce the cost of funding.
3. Allows business units to price transactions by taking into account the costs of funding.
4. Provide an appropriate benchmark for client transactions.
5. Incorporate key elements of the firm’s interest rate and liquidity risk management frameworks / capabilities.
6. Minimize the impact of interest rate volatility on accrual businesses by transferring market interest rate risk for centralized hedging.

The Concept of Funds Transfer Pricing Model



Bank Chain FTP Example



- ✓ SOFR funds are theoretically unlimited on interbank market.
- ✓ SOFR is a right benchmark for client rate pricing.

Broker Dealer FTP

FTP methodologies may differ across businesses, legal entities and geographies.

Broker Dealer entities fund their balance sheets through a combination of

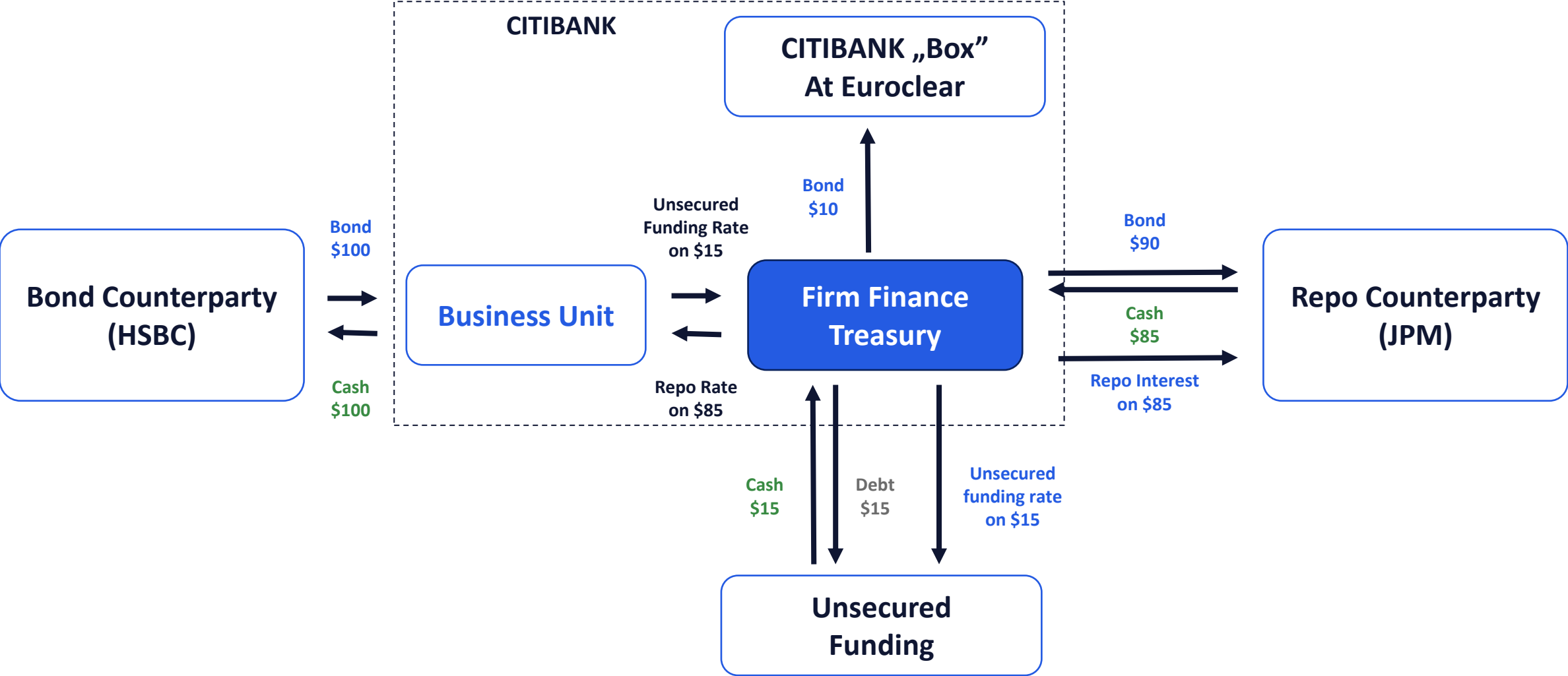
1. Secured Financing (e.g. repo and Securities Lending)
2. Internal Coverage
3. Unsecured Financing (Debt and Equity)

There are different rates associated with each of these sources of financing

- a. Secured financing rates are determined by secured financing pool calculators (based on external transactions and internal coverage).
- b. Unsecured financing is generally charged at the weighted average cost of capital.

The blend of secured and unsecured financing (both balance and rate) applied is generally driven by the ability to finance in the secured markets.

Broker Dealer FTP Example





Does anyone have
any questions or
comments?

All contributions are welcome!

EARLY CAREER OPPORTUNITIES IN HUNGARY



@Citi



@CitiCareers



@CitiCareers



@CitiCareers



SCAN THE QR CODE
TO LEARN MORE

SEARCH & APPLY FOR JOBS AT
[JOBS.CITI.COM/HUNGARY](https://jobs.citi.com/hungary)

